Gautham Ramkumar

Computer Vision & 3D Perception Engineer

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F-1 STEM OPT eligible; no immediate sponsorship required

Summary

Robotics M.S. candidate specializing in computer vision and 3D perception, from classical multi-view geometry and bundle adjustment to learned enhancement models deployed on edge hardware.

Education

Northeastern University, Boston, MA **2024 – 2026**

M.S. Robotics — Concentration in Electrical & Computer Engineering

- **Coursework:** Advanced Perception, Robot Sensing & Navigation, Autonomous Field Robotics, Mobile Robotics, Robot Mechanics & Controls, Legged Robotics

Vellore Institute of Technology (VIT), Chennai, India **2020 – 2024**

B.Tech. Electronics & Communications Engineering

Technical Skills

Languages: Python, C++, MATLAB
Perception & Vision: Visual SLAM (RTAB-Map), Stereo Vision, Depth Estimation, Object Detection (YOLO), 3D Reconstruction (SfM), Feature Matching (SIFT/ORB), Point Cloud Processing, Loop Closure Detection
Deep Learning: CNNs, RNNs / ConvGRU, U-Net, 3D Convolutions, Vision-Language Models, Transfer Learning, Loss Function Design, Edge Deployment
Frameworks & Tools: ROS2, PyTorch, OpenCV, GTSAM, Open3D, TensorRT, CUDA, COLMAP, Docker, Git
Hardware & Sensors: ZED Mini Stereo Camera, Intel RealSense D435, LiDAR, VectorNav IMU, NVIDIA Jetson Nano

Selected Projects

3D Gaussian Splatting from Scratch — 31.42 dB PSNR, 0.9531 SSIM **Jan 2026 – Mar 2026**

Python · PyTorch · CUDA · COLMAP · Docker

- Implemented the full Gaussian optimization pipeline from scratch, including densification, cloning, splitting, pruning, and adaptive parameter control, on the official EWA CUDA rasterizer.
- Established 7,500 iterations as the optimal stopping point for small scenes at 31.42 dB PSNR / 0.9531 SSIM, over 16 dB better than the 30k checkpoint.

DarkSight — Low-Light Perception — 21 FPS real-time, 17.25 dB PSNR **2025 – Present**

PyTorch · TensorRT · RAFT · YOLOv8 · RealSense

- Designed a DS3DConv encoder-decoder with a ConvGRU bottleneck for frame memory and RAFT optical-flow warping for temporal consistency.
- Tuned a perception-aware loss against YOLOv8 features and exported the model to TensorRT INT8/FP16 for edge deployment.

3D Reconstruction via Structure from Motion — ~60-70% BA cost reduction **2025**

Python · OpenCV · GTSAM · Open3D

- 'Built an incremental SfM pipeline from scratch': 'SIFT features, RANSAC outlier rejection, essential-matrix decomposition, and triangulation.'
- Optimized roughly 5,500 landmarks through GTSAM factor-graph bundle adjustment, cutting bundle adjustment cost 60-70%.

Semantic Color Constancy — +63% color accuracy **2025**

Python · PyTorch · YOLOv8 · OpenCV

- Conditioned illumination correction on semantic object identity via YOLOv8 detections and per-object chromaticity priors.
- Learned priors across 80 COCO categories from 5,600 derived images.

Experience

SmartInternz **2023**

IoT Engineering Intern

- Built real-time sensing pipelines and cloud logging systems on IoT hardware.
- Integrated embedded sensor data acquisition with cloud infrastructure.

Awards

2nd Place, Revolute Hackathon (Aug 2026) — Voice-controlled robotic manipulation for cooking, built in a weekend.